

Day 20 Problem Set

Geometry Summer Credit | Area, Composite Figures, and Coordinate Geometry

Name:	Date:	Period:
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Directions

Show the formula or decomposition you use. Label square units for area and linear units for perimeter.

Part A: Area Formulas and Composite Figures

1.	Find the area of a parallelogram with base 18 and height 11.
2.	Find the area of a triangle with base 24 and height 9.
3.	Find the area of a trapezoid with bases 15 and 27 and height 8.
4.	A 20 by 14 rectangle has a 6 by 5 rectangle removed. Find the remaining area.
5.	A figure is a 9 by 12 rectangle plus a right triangle with legs 9 and 8. Find the total area.
6.	A square with side length 18 has a circular fountain of radius 4 removed. Find the remaining area in terms of pi.
7.	Error analysis: A student finds trapezoid area by multiplying only the longer base by the height. Explain the error.

Part B: Coordinate Area and Perimeter

8. Rectangle A(2, 1), B(11, 1), C(11, 6), D(2, 6): find area and perimeter.

9. Triangle P(0, 0), Q(10, 0), R(10, 8): find area.

10. For the triangle in item 9, find the perimeter.

11. A coordinate figure fits inside a 14 by 10 rectangle. A right triangle with legs 4 and 5 and a rectangle 3 by 6 are removed. Find the area.

12. Find the distance between (-3, 4) and (5, 10).

13. A parallelogram has vertices A(0, 0), B(8, 0), C(11, 5), D(3, 5). Find its area.

14. Spiral review: A circle has radius 12 and central angle 75 degrees. Find the arc length.

Day 20 Problem Set

Continued | Mixed Review

15.	A circle has radius 9 and central angle 160 degrees. Find the sector area.
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16.	Two chords intersect inside a circle. One chord has segments 5 and 12. The other has segments 6 and x . Find x .
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17.	A secant and tangent share an external point. The secant has external length 4 and whole length 25. Find the tangent length.
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18.	A circle has center (3, -2) and radius 7. Write its equation.
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19.	Challenge: A garden is made of a rectangle 30 ft by 18 ft and a semicircle with diameter 18 ft attached to one side. Find the exact area.
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